Constraint Interaction Effects in Grammar-Guided Symbolic Regression

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Abstract

Symbolic regression systems routinely apply multiple constraints—grammar restrictions, dimensional analysis, positivity requirements, depth limits—but the interactions between these constraints are not well understood. This note proposes a simple quantitative framework to measure whether constraint pairs are independent, redundant, or synergistic, and outlines a summer project to validate it on standard benchmarks.

1 Problem

Modern symbolic regression methods use constraints such as grammar restrictions, dimensional analysis, type systems, positivity requirements, depth limits, and operator restrictions to improve search efficiency (e.g., Udrescu & Tegmark, 2020; Cranmer et al., 2020; La Cava et al., 2021).

Most existing work evaluates these constraints one at a time. In practice, systems apply several constraints simultaneously, and it is unclear how they interact:

- Do different constraints provide independent benefits?
- Are some constraints redundant, removing overlapping parts of the search space?
- Can weaker constraints combine to produce a stronger effect than expected?

There is currently no standard way to measure these interactions.

2 Motivation

Constraints affect the density of valid candidates, rejection rates, structural complexity, and search throughput (e.g., Petersen et al., 2021; Biggio et al., 2021).

Most evaluations implicitly assume that constraint effects are independent. For example, if dimensional analysis removes 80% of candidates and positivity constraints remove 70%, one might expect that applying both would leave:

$$0.2 \times 0.3 = 0.06$$

or 6% of the original space. In practice, the constraints may overlap heavily (making one redundant) or target entirely different regions (producing a stronger combined reduction). Knowing which case applies would help avoid wasted constraints and design better systems.

3 Hypothesis

Constraint effects in symbolic regression are not independent. Specifically:

1. Some constraint pairs are redundant and remove largely the same expressions.
2. Some constraint pairs are synergistic and together reduce the search space more than expected.
3. The interaction pattern is a useful predictor of actual search performance.

A secondary hypothesis is that the interaction structure between constraints is a better predictor of search efficiency than individual constraint statistics alone.

4 Related Work

4.1 Grammar-Guided Symbolic Regression

Grammar-guided approaches restrict the search space using production rules (Whigham, 1995; McDermott et al., 2012). Recent systems like AI Feynman and Deep Symbolic Regression incorporate structured priors (Udrescu & Tegmark, 2020; Petersen et al., 2021), but focus on individual grammars rather than constraint combinations.

4.2 Strongly Typed and Dimensionally-Aware GP

Strongly typed GP (Montana, 1995) and dimensional analysis (Udrescu & Tegmark, 2020) reduce invalid candidates, but their interaction with other constraints has not been studied.

4.3 Constraint Satisfaction

Constraint interaction is well-studied in constraint satisfaction and optimisation (Dechter, 2003), but these ideas have not been applied to symbolic regression grammars.

5 Proposed Framework

Let S be the unconstrained expression space. For a constraint C, define the constraint density:

$$\rho(C) = \frac{|S_C|}{|S|} \tag{1}$$

where S_C is the set of expressions satisfying C. In practice, S is infinite and must be bounded. We define S_d as the set of syntactically valid expressions up to depth d under a grammar G, and estimate $\rho(C)$ via Monte Carlo sampling of N expressions from G. The sensitivity of these estimates to the choice of d and sample size N will be examined empirically in Phase 1.

For two constraints C_i and C_j, define an interaction coefficient:

$$M(i,j) = \frac{\rho(C_i \wedge C_j)}{\rho(C_i)\,\rho(C_j)} \tag{2}$$

Interpretation:

- $M \approx 1$: constraints are independent,
- $M > 1$: constraints are redundant (overlap),
- $M < 1$: constraints are synergistic (reinforce each other).

Computing $M(i,j)$ for all constraint pairs gives an interaction matrix that summarises how the constraints relate. If constraints are highly redundant ($M > 1$), adding further constraints is unlikely to substantially reduce the effective search space. Conversely, synergistic constraints ($M < 1$) may produce larger-than-expected reductions in candidate density and therefore stronger effects on search dynamics.

6 Expected Scientific Contribution

This work seeks to determine whether search-space reduction effects combine predictably when multiple constraints are applied simultaneously.

The resulting framework may help explain why some combinations of constraints improve symbolic regression performance while others provide little benefit despite comparable reductions in feasible space.

7 Feasibility

This project uses existing symbolic regression tools and does not require building a new SR algorithm. The experiments involve expression sampling, constraint checking, and standard benchmark runs—all feasible on a personal workstation with Python (NumPy, SymPy) and an existing SR library such as PySR or gplearn.

8 Proposed Timeline

Phase 1: Literature Review and Setup (2–3 Weeks)

- Survey grammar-guided SR, typed GP, and constraint satisfaction literature.
- Pin down definitions of $\rho(C)$, $M(i, j)$.

Deliverable: Research note and experimental protocol.

Phase 2: Constraint Density Estimation (2–3 Weeks)

- Implement constraint modules (depth, positivity, dimensional, EML, operator restrictions).
- Build an expression sampling pipeline.
- Estimate $\rho(C)$ for each constraint individually.

Deliverable: Per-constraint density estimates.

Phase 3: Interaction Analysis (3–4 Weeks)

- Compute $\rho(C_i \wedge C_j)$ for all pairs.
- Construct the interaction matrix $M(i, j)$.
- Classify pairs as independent, redundant, or synergistic.

Deliverable: Interaction matrix and analysis.

Phase 4: Search Experiments (3–4 Weeks)

- Run SR on benchmark problems with individual and combined constraints.
- Measure recovery rate, rejection rate, search throughput, and evaluations-to-solution.
- Compare observed performance against predictions from the interaction matrix.

Deliverable: Experimental results and write-up.

Estimated total duration: Approximately 3–4 months (summer project).

Long-term, this work is motivated by a broader question: how do different forms of constraints reshape symbolic search spaces, and can these effects be predicted before running expensive search experiments?